Portfolio Task: Distributed Architecture Plan

System: Najm Application

1. Position Your System in the Distributed Landscape

Selected Model: Client-Server Architecture supported by a Distributed Cloud Infrastructure.
Justification: This model is highly appropriate for the Najm application. The user's mobile device acts as the "Client" that collects accident data (photos, GPS location), while the "Distributed Servers" handle the heavy processing and data storage. This architecture is ideal for handling sudden peak loads—such as during adverse weather conditions when traffic accidents spike. It allows for efficient resource sharing and prevents the system from relying on a single centralized machine that could easily become overwhelmed.

2. Define the Distributed Architecture

Nodes and Device Distribution:

- **Client Nodes:** The mobile devices of drivers and Najm investigators.
- **Load Balancer Node:** Acts as the entry point to intelligently distribute incoming traffic.
- **Application Server Nodes:** A distributed fleet of servers responsible for processing reports and executing business logic.
- **Data Nodes (Distributed Databases):** Synchronized Master and Replica databases for secure and redundant data storage.

Low Latency Data Transfer: To ensure minimal delay, data transfer will be optimized by compressing images locally on the client device before transmission. Furthermore, the system will utilize lightweight RESTful APIs and geo-routing to connect users to the closest available server node, drastically reducing latency during critical moments.

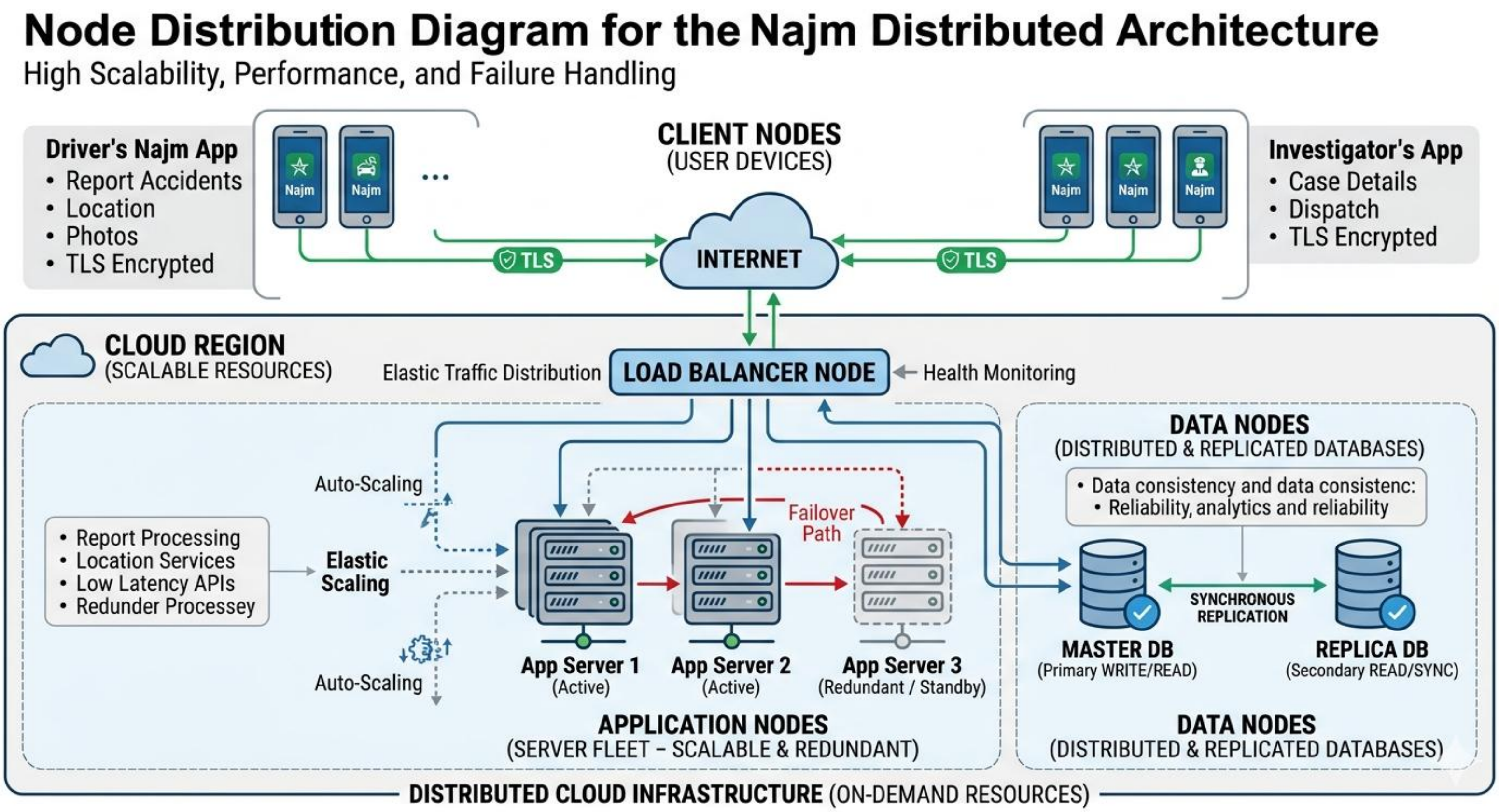
3. Reliability & Security

Failover Plan: In the event of a *Node Failure* (e.g., an application server crashes), the Load Balancer will immediately detect the unresponsiveness and reroute all new and pending requests to healthy, redundant nodes, preventing a system-wide crash. In the case of a *Network Outage* on the user's end, the mobile app will utilize an offline mode to cache the accident report locally, automatically resuming the upload process once the connection is restored.
Security Standard: The system will implement the **TLS (Transport Layer Security)** protocol. This ensures that all sensitive data in transit—including personal identities, geographic locations, and accident photos—is heavily encrypted between the client nodes and the distributed servers, protecting it from interception or tampering.

4. Industry Alignment

Cloud Computing Leveraging: This strategy heavily leverages Cloud Computing (Infrastructure as a Service). By hosting the Najm platform on a cloud provider, the system benefits from *Auto-scaling*. During peak accident hours (e.g., heavy rain), the cloud infrastructure automatically provisions additional server nodes on-demand to handle the massive influx of data. Once the peak subsides, it scales down, significantly reducing unnecessary infrastructure costs while guaranteeing continuous resilience.

5. Node Distribution Diagram



Continuity and Resilience: As illustrated in the diagram, this distributed architecture explicitly supports continuity by eliminating any Single Point of Failure (SPOF). The redundancy across application servers and databases ensures that the Najm app remains highly available to drivers in distress, fulfilling its real-world critical mission seamlessly.